# Where Do Bikes Go
# Analyzing Bike Share Commuting Patterns

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## Objective

Identify Capital Bikeshare stations that consistently have more departures than arrivals, and analyze how these imbalances differ between weekdays and weekends. We will produce ranked lists of persistently imbalanced stations that may warrant further investigation for bike redistribution. Trip records alone cannot show actual bike availability or dock capacity, or establish where redistribution would have the greatest impact.

## Motivation

Bike-share operators move bikes between stations to maintain service. Understanding which stations consistently lose bikes through recorded trips, and when, can help inform redistribution planning. This project connects to course themes of data retrieval, cleaning, integration, and pattern recovery from imperfect records. It will produce a reproducible analysis of recurring station-level flows.

## Data

We will download Capital Bikeshare public trip history files as monthly CSVs from https://capitalbikeshare.com/system-data. The study will cover three consecutive months, with an estimated 300,000 to 1,000,000 trip records. We will confirm the study period and actual record counts during initial inspection; the intended volume exceeds the 50,000-record target.

Expected challenges include inconsistent station identifiers or names, missing station information or coordinates, and possible schema or timestamp differences across files. Missing locations will be flagged rather than assumed to indicate lost bikes. We will document exclusions and retain valid station endpoints where appropriate. Dock capacity, real-time availability, and weather data are outside the planned dataset and will limit interpretation.

## Approach

**1. Retrieve and store.** Preserve the raw monthly files and combine them in DuckDB for efficient querying, recording file sources and record counts.

**2. Clean and standardize.** Parse timestamps, standardize station identifiers and names, check duplicates and invalid coordinates, and document missing-value handling. Produce a data-quality summary with counts before and after cleaning.

**3. Calculate imbalances.** Count departures and arrivals by station, hour, and day of week. Define net flow as arrivals minus departures, so negative values indicate net departures. Compare average flows per weekday and weekend day rather than unequal raw totals.

**4. Assess persistence.** Compare station profiles across the three months, identify recurring net departures, and investigate anomalies. Describe morning-origin, evening-destination, or weekend-oriented patterns without treating trip purpose as known.

**5. Visualize and validate.** Create ranked station lists, hourly profiles, heatmaps, and weekday/weekend comparisons. Reconcile aggregate event counts with retained trip endpoints and review selected station calculations.

Tools: Python with pandas, DuckDB, matplotlib/seaborn, Jupyter notebooks, and Git.

## Team Responsibilities

**Brett Dowling — Data acquisition and preparation.** Download and organize the monthly files, integrate them into a consistent dataset, validate schemas, standardize station identifiers, and document missing or invalid values. Deliver the reproducible cleaning pipeline and data-quality summary. Lead the Data section and the retrieval and cleaning portions of the Approach.

**Devang Gupta — Imbalance metrics and temporal analysis.** Calculate station-level flows, compare weekday and weekend averages, and evaluate consistency across months. Deliver documented analysis code, station rankings, and a findings summary. Lead the Objective section and the metrics and analysis portions of the Approach.

**Camden Goering — Visualization and interpretation.** Develop reproducible heatmaps, hourly profiles, and comparison charts; examine selected stations and exceptions; and explain implications and limitations. Lead the Motivation and Expected Outcome sections, the visualization portion of the Approach, and organization of the final report and presentation.

**Shared responsibilities.** Finalize the study period, definitions, scope, and milestones together. Each member writes the final report material for their own work and contributes to editing and presentation preparation. Devang reviews data preparation, Camden reviews analysis results, and Brett reviews visualizations and interpretation.

| Week | Milestone |
|---|---|
| 1 | Finalize scope and study period; download files; inspect schemas; set up the repository and database. |
| 2 | Complete the cleaning pipeline and data-quality summary; document missing-value handling. |
| 3 | Compute hourly imbalances, validate counts, and produce initial station rankings. |
| 4 | Compare weekday and weekend patterns across months; draft visualizations. |
| 5 | Investigate anomalies, refine interpretations, and assemble the report draft. |
| 6 | Complete peer review; finalize code, visualizations, report, and presentation. |

## Timeline and Milestones

## Expected Outcome

The final result will be a reproducible data analysis with ranked lists of persistently imbalanced stations, separate weekday and weekend comparisons, hourly profiles, and heatmaps showing when net departures and arrivals peak. Deliverables will include a cleaning pipeline, analysis and visualization notebooks, and a report explaining methods, findings, limitations, and potential redistribution implications for a general audience.

## Scope

The project is limited to descriptive flow analysis over three months, with no real-time modeling or causal claims. Its substantial data volume, integration of monthly files, documented cleaning decisions, temporal

aggregation, and cross-month validation go beyond a few basic charts. Success will be evaluated through reproducibility, reconciled counts, and clear evidence for recurring patterns and their limits.